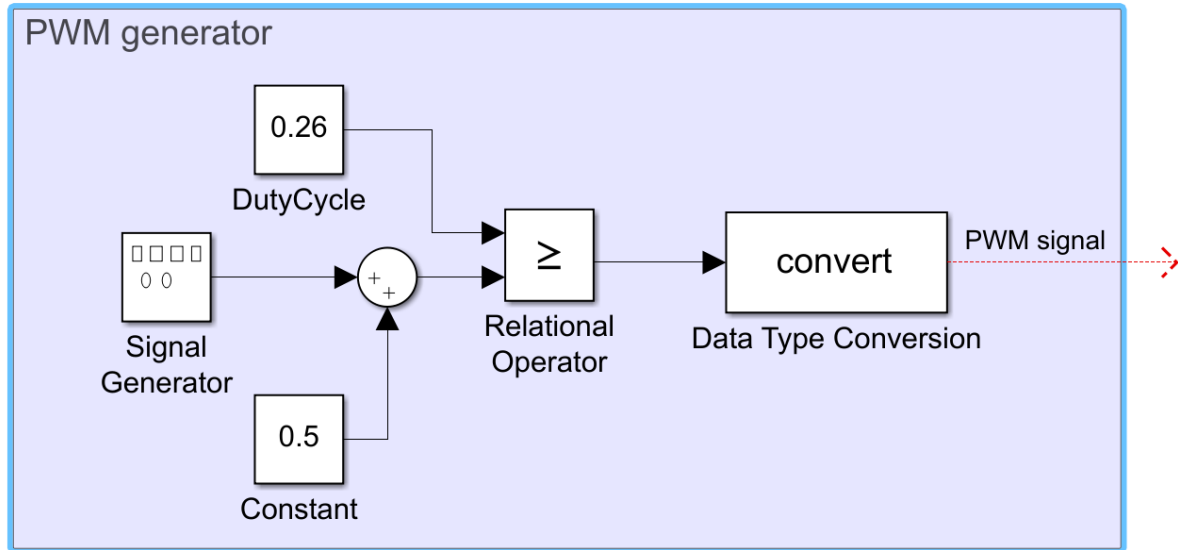


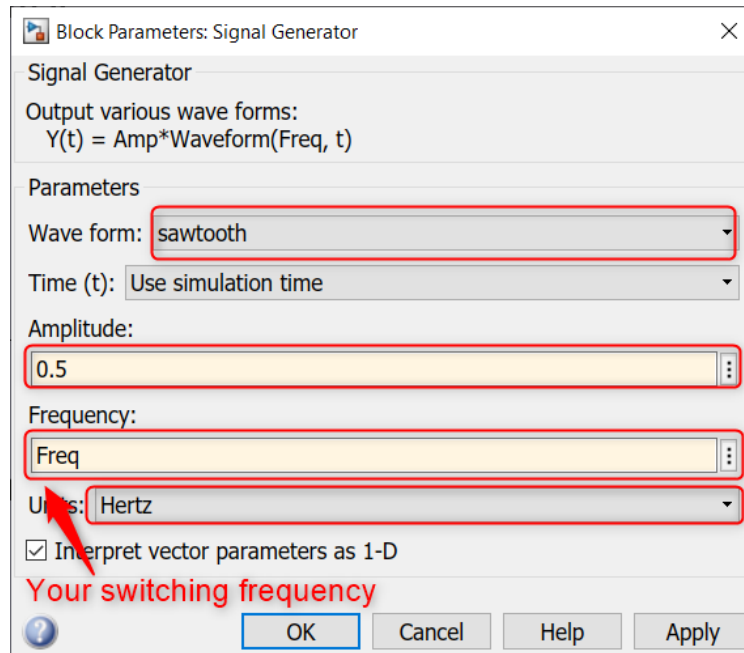
Annex 1: Simulations hints

a) PWM generation

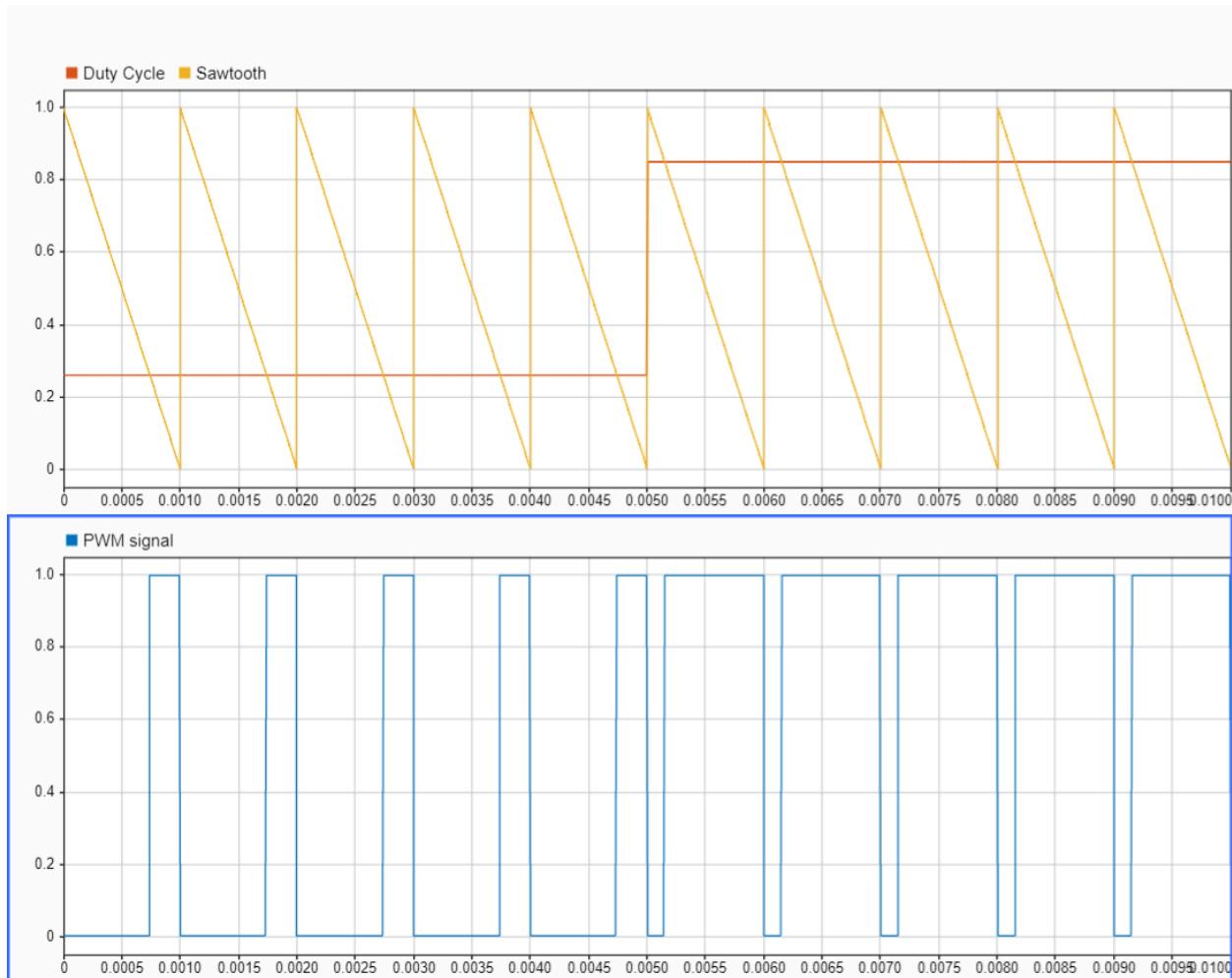
Instead of using the PWM block offered by Simulink, the recommended way of generating PWM signal is the following: generate a triangular/sawtooth waveform with the desired frequency, and compare it with the duty cycle, generating the PWM signal.



The settings of the Signal generator:



With these settings, the output of the signal generator will be a sawtooth signal in range of $[-0.5, 0.5]$. An offset of 0.5 is applied to shift the range to $[0, 1]$. With the Relational operator block, a square wave is generated, whose duty cycle is changing with the constants value.



b) Simulation step time

The default setting for Simulink files allows the solver to adapt the integration step time, which can cause improper simulation, with deviation from reality. To improve this, we can choose a fixed time solver, with a low enough simulation step time, to preserve the details and dynamics of the system.

For this project, the recommended step time is $1e-6$ (1 μ s). The length of the simulation can be limited to **0.05** (50ms). This is enough to observe the step response of the system, with the PWM filtration.

To open the setting window, press **Ctrl+E**.

Configuration Parameters: SimpleModel/Configuration (Active)

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Simulation time

Start time: 0.0 Stop time: 0.05

Solver selection

Type: Fixed-step Solver: auto (Automatic solver selection)

▼ Solver details

Fixed-step size (fundamental sample time): 1e-6

Tasking and sample time options

Periodic sample time constraint: Unconstrained

- ☐ Treat each discrete rate as a separate task
- ☐ Allow tasks to execute concurrently on target
- ☐ Automatically handle rate transition for data transfer
- ☐ Higher priority value indicates higher task priority

OK Cancel Help Apply